

# Final Project

## Time Series Analysis

Conduct a complete applied time series study in a real operational context where at least three related series are monitored. Treat the series both as individual processes and as components of a system, producing forecasts, diagnostics, and interpretable evidence in a professional or research setting. The project should result in two deliverables, the code with models and interpretations, and an interactive dashboard implemented in Streamlit that allows a practitioner to explore the data, reproduce forecasts for a selected horizon, and inspect model diagnostics. The narrative of the project should move from the business or scientific question, through data preparation and exploratory analysis, into model identification, estimation, validation, and finally forecast communication.

The chosen context must be documented with enough detail for a reader outside the team to understand why the forecasting exercise matters and what decisions the forecasts are intended to support.

The analysis must compare, at a minimum, three families of models covered during the semester. The first is exponential smoothing, including simple, Holt, and Holt-Winters variants. The second is the Box-Jenkins methodology, comprising AR, MA, ARMA, and ARIMA. The third is the machine learning family, which includes tree-based models and one neural network.

The Streamlit application should expose, at a minimum, a selector for the target series, a control for the forecast horizon, a visualization of the historical data with overlaid forecasts and prediction intervals, a panel with residual diagnostics, and a summary table comparing accuracy across the implemented models. The dashboard is evaluated on clarity and on its fitness as a communication tool for a non-technical decision maker, not on visual ornamentation.

The potential data sources include energy and utilities, macroeconomics and finance, public health and epidemiology, transportation and mobility, retail and demand forecasting, environment and climate, and tourism and hospitality. Students may propose alternative sources provided that the data are publicly accessible.

Students submit a public repository containing a reproducibility notebook and a link to the deployed Streamlit dashboard. Other libraries and deployment platforms are accepted, such as Shiny (R), Google Cloud, AWS, and Azure. A 5 minutes max video is required to present the results. In the video, do not explain the code; focus on the project's motivation, results, and impact in the selected context.